Programmable Heat-Exchanger Fouling, Duty and Cleaning-Screening Models

Independent mathematical assessment of LMTD, counterflow effectiveness–NTU, thermal constraints and transparent cleaning scores

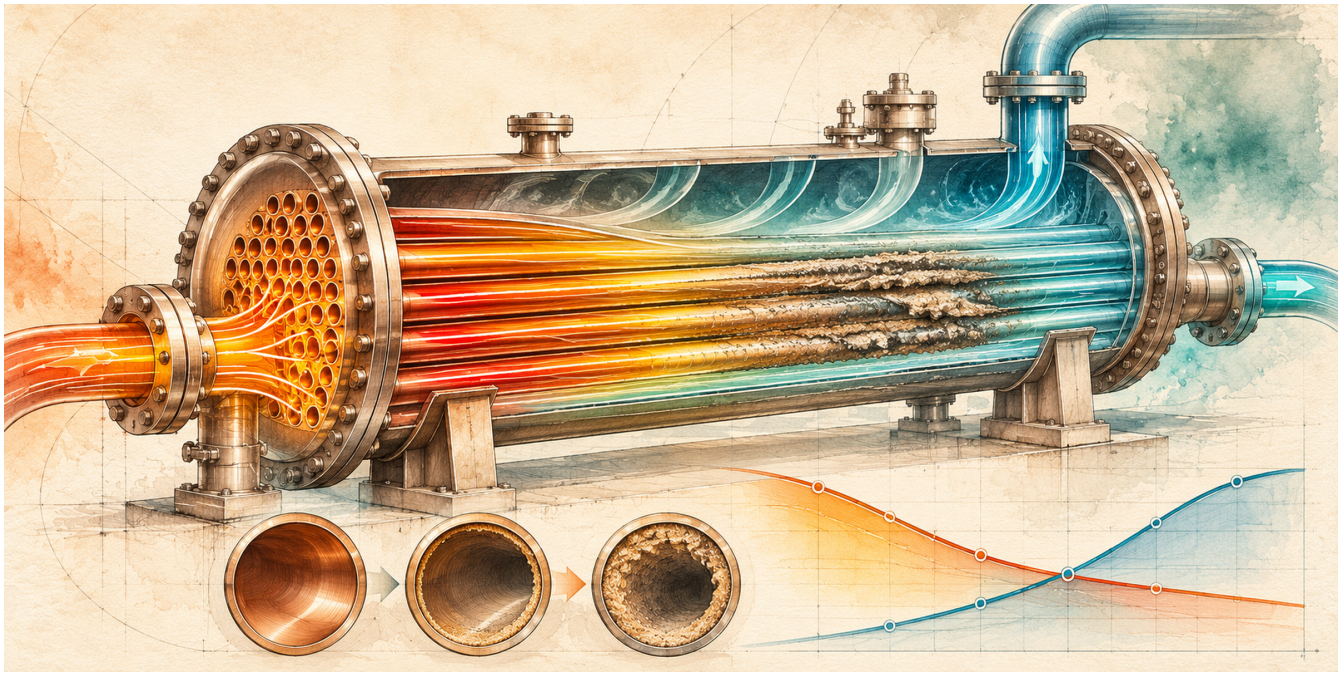


Figure 1. Concept illustration of a shell-and-tube heat exchanger and deposit accumulation. Not a computed research output.

Functional research demonstrator | 14 November 2026 | English report

Actual remote mathematical graph execution: paceval MCP 4.26. Independent high-precision reference validation. Synthetic engineering inputs only. Local SDK C++ runtime execution remains unverified pending an ABI-compatible library.

1. Industrial problem and the research question

Industrial heat exchangers transfer energy from a hot process fluid to a cooler stream through a wall. During operation, mineral scale, biological deposits, corrosion products and process residues can accumulate on that wall. The deposits add a thermal resistance and often complicate operating decisions. An exchanger can become duty-limited, yet the observed duty may also change because inlet temperatures and flow conditions change, even if the fouling state does not.

A typical maintenance engineer therefore needs to distinguish an actual thermal constraint from an apparent loss caused by a changed operating point. Temperature approaches, heat-capacity rates, pressure-drop observations and measurement credibility are important inputs. Decisions about cleaning additionally depend on plant shutdown plans, cleaning effectiveness and explicit financial assumptions.

Concrete hypothetical operating example

Consider a steady counterflow exchanger in a process heat-recovery loop. The hot inlet temperature and a chosen overall fouling resistance vary across an operating map, while the cold inlet temperature, heat-transfer area and the two fluid heat-capacity rates remain fixed. For each state, the demonstrator produces an energy-consistent duty estimate, two outlet temperatures, an approach-temperature margin, a duty margin and an illustrative cleaning score.

The scientific question is narrowly formulated: Can the real paceval execution engine evaluate all mathematically explicit stages of these linked heat-exchanger models as reusable persistent graphs, with independent verification and traceable change of model version? This is not a test of general-purpose heat-exchanger design software, industrial control certification or realized cleaning savings.

Why two model paths?

The original WP 15 proposal specifies an effective conductance, an LMTD duty expression and a simplified cleaning cost. These are useful, but the terminal temperature differences in LMTD are independent input observations. They cannot automatically be treated as self-consistent outlet predictions. The present work preserves that version and adds a fully coupled ideal counterflow model using the established effectiveness–NTU method.

2. Scientific and industrial state of the art

The ASHRAE Fundamentals Handbook (2025, Chapter 4) describes both logarithmic mean temperature difference and heat-exchanger effectiveness–NTU calculations. The familiar LMTD relation uses the two local terminal temperature differences when the configuration and overall coefficient are defined. The epsilon–NTU formulation is appropriate when thermal capacity rates, heat-transfer conductance and inlet temperatures are known. IDAES offers an established process-systems framework with a dedicated NTU exchanger unit model.

Industrial fouling monitoring is not confined to thermal duty. Studies in 2022 and 2023 evaluate joint thermal and hydraulic indicators because heat duty itself varies with upstream conditions. A 2025 taxonomy distinguishes fouling mechanisms and operational definitions. In 2026, work on data-driven plate-exchanger health monitoring and a systematic review of cleaning-in-place models emphasize sensor data, dynamic conditions and model limitations. These research advances provide context, not validation of the present synthetic model.

Practical ecosystem and boundaries

Engineering workflows may use manufacturer and TEMA-based design practices, process-simulation environments, thermal instrumentation, inferred heat-transfer coefficients and cleaning-history databases. Physics-based and machine-learning models can help estimate the fouling trajectory, whereas a static heat-duty expression only evaluates a specified state. The original paceval model is most credible as an explicit, inspectable calculation layer in a larger workflow that continues to own plant data, hydraulics and the final maintenance decision.

The current study does not assume a universal fouling coefficient or a known deposit growth law. The listed fouling resistance values are synthetic research inputs. Two nominal pressure-drop quantities are externally provided; this study does not infer them from fouling. No claims about particular industrial facilities, energy savings, uptime or customer outcomes are made.

Research sources

ASHRAE 2025 Fundamentals Ch. 4; IDAES NTU Heat Exchanger (current stable documentation); Patil et al., Chemical Engineering Research and Design (2022); Performance monitoring of heat exchanger networks (2023); Augustin, Melo and Wilson (2025); MLP-based plate-exchanger health monitoring (2026); Madhu et al., Journal of Food Science (2026). Full links appear on the final page.

3. Physical variables, units and domains

Variable	Definition / unit
U_c, R_f	Clean overall coefficient (W/m²/K), total fouling resistance (m² K/W)
A, F_c	Heat-transfer area (m²), geometry correction factor (dimensionless)
$T_{h,in}$, $T_{c,in}$	Hot and cold inlet temperatures (°C), positive hot-to-cold inlet difference
C_h, C_c	Mass flow times heat capacity, W/K
ΔT_1, ΔT_2	Externally specified, positive terminal differences (K) for v1
R_{remain}, Q_{target}	Residual fouling after hypothetical cleaning (m²K/W); target heat duty (W)
H, Price, C_{clean}	Assumed evaluation hours, thermal shortfall valuation (EUR/kWh), cleaning cost (EUR)
Approach, ΔP, ΔP_{max}	Minimum terminal approach (K); externally supplied and limiting pressure drops (Pa)

Illustrative synthetic input envelope

The main 49-point map uses seven hot inlet temperatures from 95 to 155 °C and seven prescribed fouling resistances between zero and 0.0012 m² K/W. The remaining research constants are a cold inlet of 35 °C, Ch=15,000 W/K, Cc=12,000 W/K, clean coefficient Uc=950 W/m²/K and area A=32 m². The synthetic target is 550 kW with a specified minimum approach of 5 K. The v1 terminal gradients are independently specified as 45 and 25 K.

The domain checks reject inverted inlet temperatures, nonpositive capacity rates, negative fouling resistance, invalid terminal temperature differences and disallowed nonfinite values. A separate stable LMTD series handles coinciding terminal differences. A separate effectiveness formula handles exactly equal capacity rates. The main batch deliberately avoids the direct-expression singular branch.

4. Original model v1: terminal LMTD duty

Effective conductance:

$$Ueff = 1 / (1/Uclean + Rf)$$

Logarithmic mean temperature difference:

$$LMTD = (\Delta T1 - \Delta T2) / \ln(\Delta T1/\Delta T2)$$

Terminal-data screening duty:

$$Qv1 = Ueff \cdot A \cdot Fcorr \cdot LMTD$$

The expression requires both terminal temperature differences to be positive, and the direct quotient has a removable singularity at $\Delta T1=\Delta T2$. Its analytic limit is $\Delta T1$. The project therefore includes an additional series graph in $u=(\Delta T1-\Delta T2)/(\Delta T1+\Delta T2)$, using $LMTD\approx(\Delta T1+\Delta T2)/2 \times (1 - u^2/3 - 4u^4/45 - 44u^6/945)$. The host chooses the branch; paceval performs the mathematical series evaluation.

The original proposal's LMTD formula remains explicitly implemented, rather than being replaced silently. This version is useful for algebraic sensitivity screening when the terminal differences are genuinely measured or externally specified. It is not a closed thermal balance unless the terminal temperatures themselves are consistent with the corresponding heat-capacity rates and heat flow.

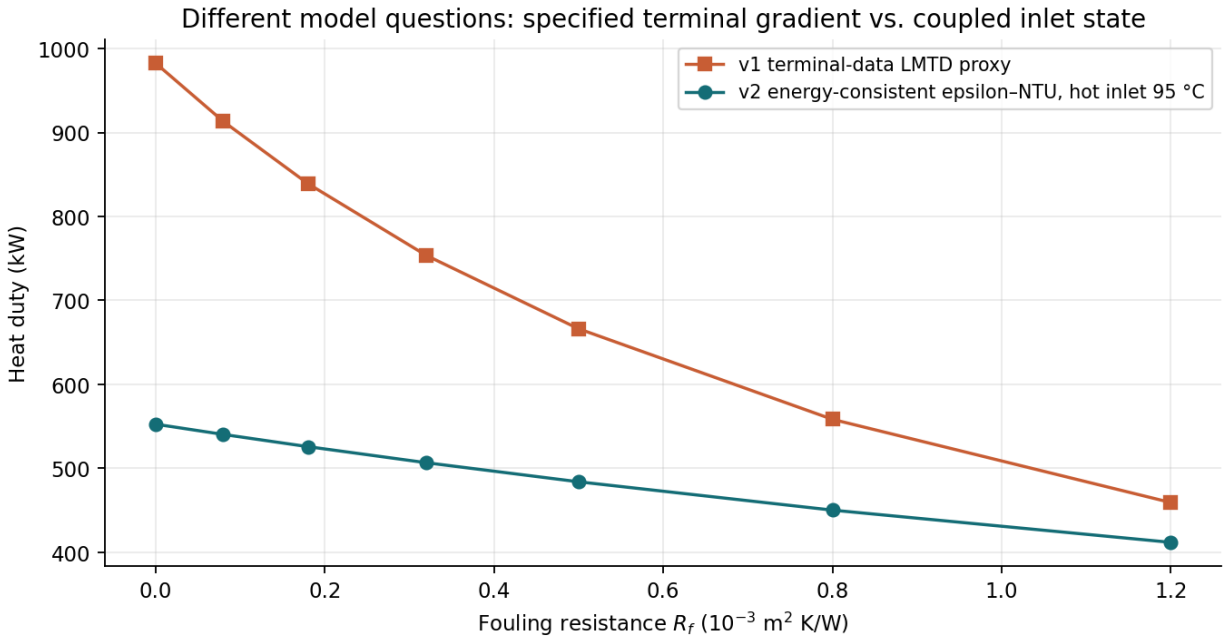


Figure 2. Both curves use actual paceval results, but their boundary assumptions differ. Their separation is not a numerical error metric or a model-accuracy ranking.

5. Extended model v2: coupled counterflow duty

The extended model assumes an ideal steady, single-phase, adiabatic counterflow exchanger with uniform effective conductance and constant heat-capacity rates. It defines:

$$C_{min} = \min(C_h, C_c);\quad C_r = C_{min}/C_{max};\quad NTU = U_{eff}\cdot A/C_{min}$$

$$z = \exp(-NTU\cdot(1-C_r));\quad \varepsilon = (1-z)/(1-C_r\cdot z)$$

$$Q_{v2} = \varepsilon\cdot C_{min}\cdot(T_{h,in}-T_{c,in})$$

The two outlet temperatures follow from energy conservation: $T_{h,out}=T_{h,in}-Q_{v2}/C_h$ and $T_{c,out}=T_{c,in}+Q_{v2}/C_c$. The minimum terminal temperature approach is the smaller of $T_{h,in}-T_{c,out}$ and $T_{h,out}-T_{c,in}$. Separate paceval graphs calculate each of these quantities, the duty margin $Q_{v2}-Q_{target}$, the approach margin, a complete energy-balance residual and a combined logical guard. Host C++ interprets the resulting 0/1 guard without granting industrial approval.

The unequal-capacity expression is not blindly applied at $C_r=1$. An independent graph evaluates the analytic balanced limit $\varepsilon=NTU/(1+NTU)$, and the host selects this branch when capacity rates are equal. Near-equal capacity rates warrant further conditioning studies; a pointwise function evaluation alone is not a proof of uniform stability.

To study fouling sensitivity, another paceval graph contains the independently derived derivative of Q_{v2} with respect to R_f. In the declared nondegenerate domain this derivative is negative: increasing the fouling resistance decreases the modeled thermal duty at fixed boundary conditions.

6. Graph inventory and execution architecture

The executed research graph set has 24 registered, persistent mathematical expressions. All use the same declared 18-variable layout so multiple independent scenarios can be evaluated through the real paceval MCP multi-graph batch method. The original local SDK C++ program is organized to use matching formulas and SDK batch calls.

Group	Graph examples and role
Thermal resistance / LMTD	Ueff; LMTD_direct; LMTD_series; Q_LMTD_v1; Q_LMTD_series_v1
Capacity and ε-NTU	capacity_ratio; NTU_v2; effectiveness_v2; Q_NTU_v2; balanced_effectiveness
Energy and thermal constraints	hot_out_v2; cold_out_v2; pinch_min_v2; duty_margin_v2; pinch_margin_v2; energy_balance_residual_v2
Cleaning option and decisions	Q_clean_option; wait_cost_score; clean_cost_score; cleaning_advantage_score; cleaning_guard; operation_guard
Sensitivity	sensitivity_U_Rf; sensitivity_Q_Rf

The execution pattern is: validated synthetic inputs → named persistent paceval graph handles → batch and scalar mathematical evaluations → host status interpretation and file output → independent high-precision verification. Domain validation, I/O, graph ownership and file-system decisions remain with the host. All core mathematical scores and margins are graph expressions, not native calculations hidden behind a trivial paceval call.

The public `paceval_main.h` and original `paceval_main.cpp` from the existing SDK are retained in src/. The local SDK bridge and demonstration source pass a C++17 syntax check. The compatible Linux runtime shared object is not available in this execution environment, so local link, CTest and execution results are expressly pending. Actual remote MCP computation results must not be represented as SDK runtime results.

7. Research design and result provenance

The main study includes 49 synthetic operating points, each evaluated with 24 persistent, distinct formula graphs. The remote paceval MCP 4.26 response reported 49 calculations per graph and no errors in the main set. This is 1,176 successful graph-point evaluations, not 1,176 independent scenarios, not a throughput measurement and not a substitute for model calibration.

A selected subset of actual returned graph results was transcribed to twelve significant decimal digits and archived as CSV. The source data are deliberately identified as transcribed observations, rather than an unmodified high-precision binary export. An independent 90-digit mpmath model re-derives the equations and confirms the 147 selected numerical values within the declared transcription-aware tolerance; 98 boolean reference comparisons pass exactly. Detailed numerical differences are retained only in technical audit files.

Two concrete observed operating points

Synthetic case	Qv2 / margin / interpretation
Th,in=95 °C; Rf=0	Qv2=552.509 kW, just above the assumed 550 kW target; model operation guard true
Th,in=95 °C; Rf=0.0008 m²K/W	Qv2=450.133 kW, below the target; operation guard false; hypothetical cleaning-advantage score positive
Th,in=115 °C; Rf=0.0012 m²K/W	Qv2=548.986 kW, just below target; operation guard false

The declared target, cleaning expense, horizons and capacities are modeling assumptions. These example outcomes do not establish plant setpoints, economic savings or operational recommendations.

8. Observed duty loss and capacity envelope

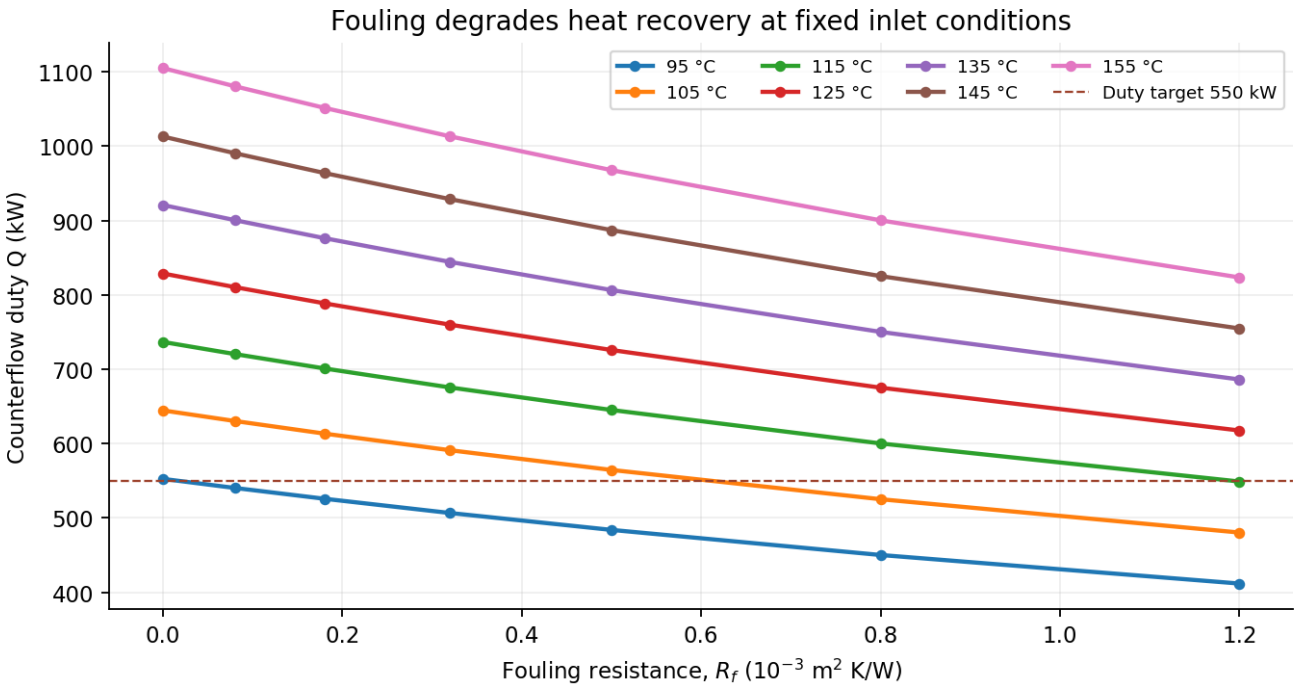


Figure 3. Actual paceval MCP model v2 heat-duty outputs versus imposed fouling resistance at seven synthetic hot inlet temperatures.

As prescribed thermal resistance rises, the effective conductance and counterflow duty decrease. At higher hot inlet temperatures, the thermal driving potential increases. These model trends are consistent with the static transfer-resistance formulation; this is not an experimentally identified fouling growth trajectory.

The main duty requirement is crossed at different resistances for different inlet conditions. This illustrates why a naive single cleaning threshold based only on elapsed time, or on an uncorrected outlet temperature, can be misleading. Pressure measurements and changes in flow distribution still require a separate hydraulic model or plant observation.

9. Two-dimensional operating maps

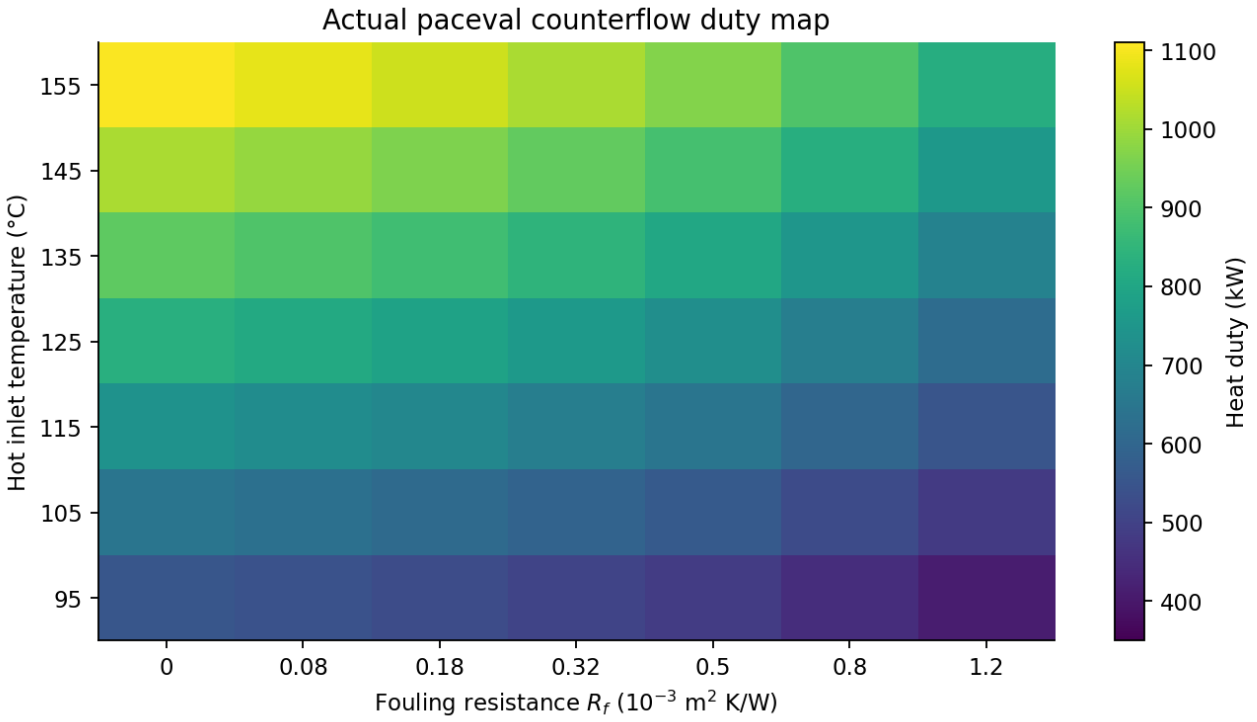


Figure 4. Heat duty (kW) computed by the actual paceval MCP 4.26 v2 graph across the 49 synthetic corner points.

The two-dimensional map makes the combined dependence on temperature driving force and fouling resistance explicit. It can be used as an interpretable mathematical screening layer after host-side unit validation and before an engineering decision. It is not a calibrated forecast for any real exchanger.

10. Constraint outcomes and cleaning model

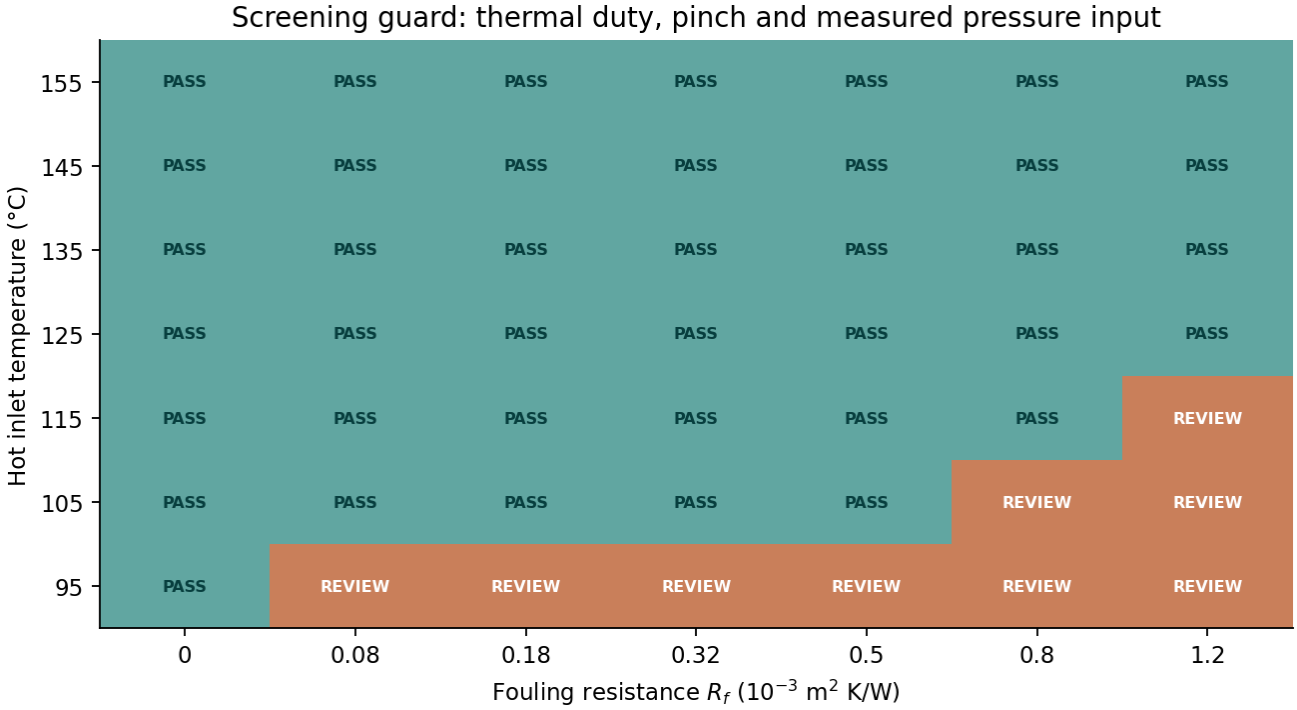


Figure 5. Mathematical screening outcomes from the actual combined paceval operation_guard graph, not a process safety or regulatory approval.

Under the declared constants, 40 of the 49 states satisfy the combined duty, pinch and externally supplied pressure-drop conditions; nine do not. Both the pass/fail and the signed continuous margins are available as paceval graph results. The model guard respects equality at the specified threshold because its expressions use greater-than-or-equal and less-than-or-equal comparisons.

11. Illustrative cleaning score and decisions

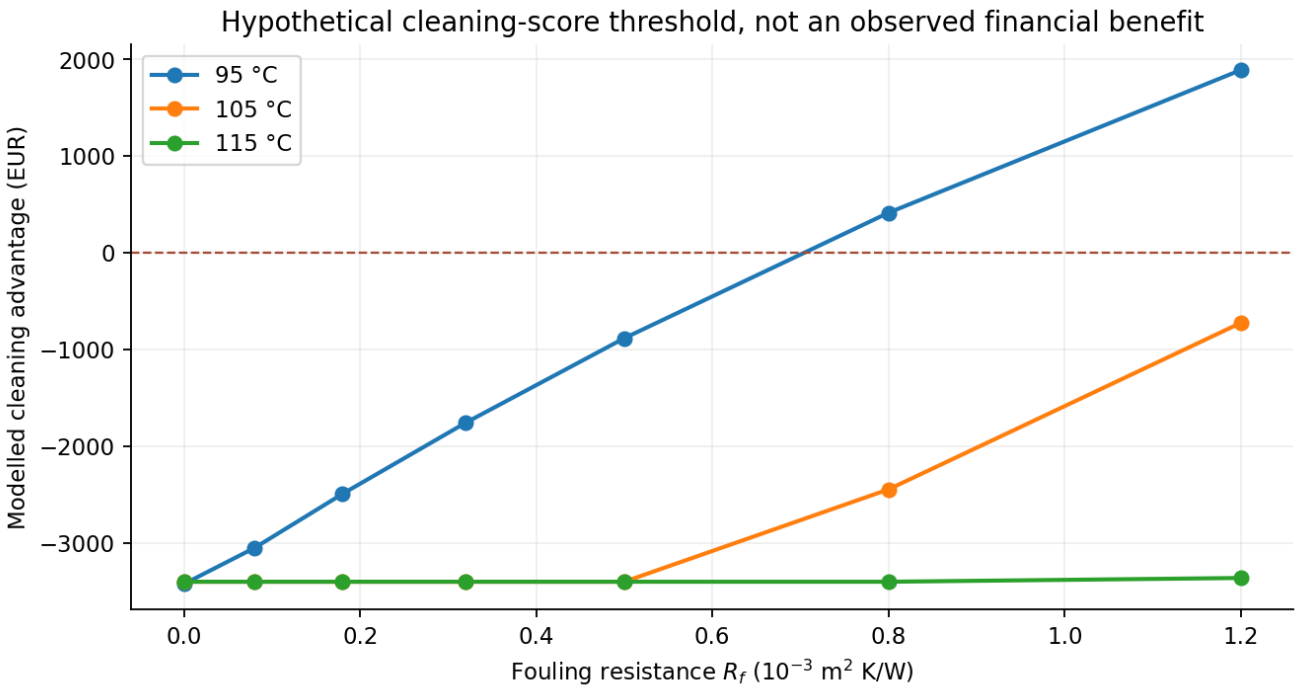


Figure 6. Real paceval MCP cleaning-advantage graph outputs. Positive values indicate a favorable hypothetical score for the declared assumptions, not observed cost savings.

For an illustrative evaluation horizon of 160 hours, a unit shortfall valuation of EUR 0.24/kWh and a lump cleaning expense of EUR 3,400, the cleaning score is the modeled difference between two option costs. The alternative post-cleaning resistance is assumed to be 0.00002 m²K/W. Only 2 of 49 operating points produce a positive cleaning guard. These parameters are not inferred from any industrial maintenance records.

A production decision would require additional downtime, chemical use, lost production, post-cleaning fouling persistence, uncertainties and operational restrictions. This research demonstrates a traceable programmable graph, not a verified cleaning schedule.

12. Numerical verification, edge cases and TINC

The original paceval MCP output was independently checked by high-precision equations implemented without copying the paceval formula strings. The checks cover the coupled thermal duty, original terminal-data duty and illustrative cleaning score for every archived input, and both logical guards. This is finite-sample numerical verification, not global formal verification.

Limits that matter

The direct LMTD quotient becomes the undefined expression 0/0 at identical terminal differences, although the mathematical limit exists. A separate paceval series expression provides 30 K at $\Delta T1=\Delta T2=30$ K and 30.00000015 K at a closely spaced test. Its non-interval point evaluation succeeded. A separate extremely near-equal TINC trial returned error code 145, and is therefore recorded as an open case, not a passed interval test.

Five actual TINC outputs for the v2 heat-duty graph were independently checked: the high-precision reference for the exact binary64 input values lies within each of the *printed decimal* lower/upper bounds. These five successful pointwise checks do not imply a global interval inclusion theorem for all possible temperatures, fouling levels or uncertain material parameters.

Additional operator tests used actual paceval graphs for signed % versus mathematical mod, infix min/max, and explicitly parenthesized relational and boolean expressions. All 21 tested graph-point combinations returned the declared 0/1 and numeric semantics without API errors. Local original-SDK converter and operator CTests still require the compatible Linux library.

13. Scientific status, reproduction and next study

The project provides a real remote paceval mathematical graph functionality demonstration for two linked modeling approaches. The independently checked synthetic results support mathematical feasibility of a reusable formula execution layer. They do not demonstrate increased speed, measured cost reduction, a true fouling estimator, calibrated plant performance, engineering signoff or regulatory approval.

Source reproduction uses the exact `src/` formula manifest, `src/heat_exchanger_paceval.cpp`, original `src/paceval_main.h` and `src/paceval_main.cpp`. A compatible Linux library must be supplied using `-DPACEVAL_RUNTIME=/absolute/path/to/libpaceval_linux_sharedLIB.so`. The public package deliberately excludes this external shared object. Without it, CMake linking and CTest cannot be certified as passed.

For a meaningful industrial follow-up, collect licensed temperature/flow/pressure histories and cleaning events; independently calibrate clean-state conductance and thermal properties; distinguish hydraulic fouling from thermal resistance; quantify identifiability; validate on held-out periods; and measure how formula revisions affect engineering maintenance workflows. An online deployment should be considered only after separate process-safety and operator governance assessments.

Principal references

• ASHRAE. 2025 Fundamentals Handbook, Chapter 4: Heat Transfer. https://handbook.ashrae.org/Handbooks/F25/SI/f25_ch04/f25_ch04_si.aspx

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• paceval Mathematical Engine official API. https://paceval.com/api/.